Text as Data

Class 3: The Federalist Papers as a case study

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Where we are in the course

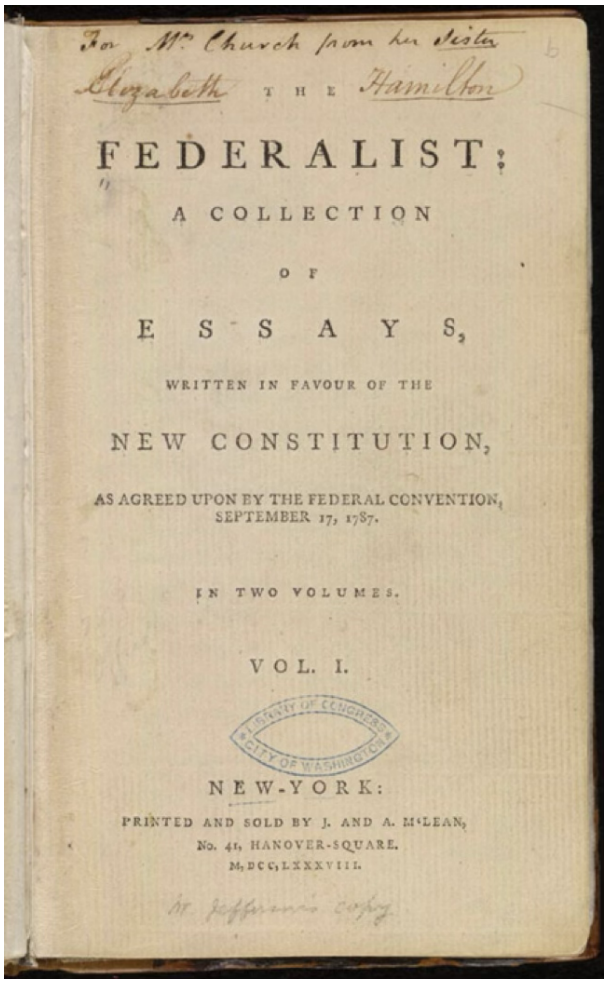
- Class 1: text becomes a structured object: documents, tokens, vectors
- Class 2: counts, rates, multinomial likelihoods, log odds, and preprocessing
- Today: use those tools on a famous authorship problem

The Federalist Papers

- Eighty-five essays published in 1787–1788 under the name *Publius*
- Written by Alexander Hamilton, James Madison, and John Jay
- Twelve essays (49–58 and 62–63) are historically disputed; the supplied data label paper 58 as Madison
- Our question: can regularities in language help distinguish Hamilton and Madison?



From newspaper essays to a book



The essays first appeared under the shared pseudonym *Publius*. They were later collected as a book, which is part of why the authorship puzzle became a durable object of study.

The historical object is therefore already a collection: many documents, several authors, and a disputed subset.

1788 collected edition.

What the authorship question is (and is not)

Authorship classification asks whether a document resembles texts attributed to one author more than another.

It does not recover a hidden author with certainty. It compares a representation, a reference set, and an explicit rule.

A useful historical benchmark

Mosteller and Wallace's study made the disputed essays a benchmark for statistical authorship analysis.



Their original work used word frequencies and probabilistic discrimination.

Our model is deliberately smaller and more transparent. It illustrates the workflow, not a replication of the original analysis.

Mosteller and Wallace (1963), "Inference in an Authorship Problem," *JASA*, doi:10.1080/01621459.1963.10500849

The computational setting changed



The original study was conducted when text analysis meant punch cards, mainframes, and carefully selected word lists.

The core problem is still familiar: define a measurable representation, compare documents, and show how the result depends on the choices.

The data file

- One row per essay; the body is in the `text` column
- Metadata include the paper number and a source author label
- Text is lowercased and tokenized with the rule `re.findall(r"[a-z]+", text.lower())`

Use the dataset labels as supplied

Analysis group	Papers	Tokens	Use
Hamilton reference	51	113,078	reference profile
Madison reference	15	40,998	reference profile
Jay reference	5	8,430	descriptive comparison
Joint attribution	3	5,696	excluded from H–M training
`HAMILTON OR MADISON`	11	22,020	held out for the case study

All analysis groups follow the `author` column in `federalist.csv`. Paper 58 is therefore part of the Madison reference set in this exercise.

Preprocessing is part of the measurement

- Tokenization decides what can be counted
- Including the shared greeting or `PUBLIUS` adds volume but little authorship information
- Removing stopwords can erase precisely the function-word signal that often carries style
- Stemming, spelling normalization, and footnote removal would define a different measurement

Why function words?

Function words such as *by* and *upon* are frequent, grammatical, and less tied to a particular topic than nouns.

Man is a content word here. It is useful as a contrast because it can move with subject matter and argument.

The feature choice is a substantive decision: it says what kind of authorship signal we are trying to measure.

Pooled counts are not yet comparable

Reference set	tokens	man	by	upon
Hamilton	113,078	104	859	374
Madison	40,998	17	474	7

Raw counts mostly reflect how much text we have. Rates answer a different question:

$$p(w \mid a) = \frac{\text{count of } w \text{ in } a}{\text{all tokens in } a}.$$

Three-word profiles, conditional on selection

Reference set	$q(\text{man})$	$q(\text{by})$	$q(\text{upon})$
Hamilton	0.077786	0.642483	0.279731
Madison	0.034137	0.951807	0.014056
Paper 49	0.117647	0.882353	0.000000

Here q conditions on one of the three selected words occurring, so every row sums to one. Paper 49 is closer to Madison on this small profile, but similarity is a descriptive result, not a posterior probability of authorship.

From counts to a vector space

Represent each text as a three-dimensional conditional three-word vector:

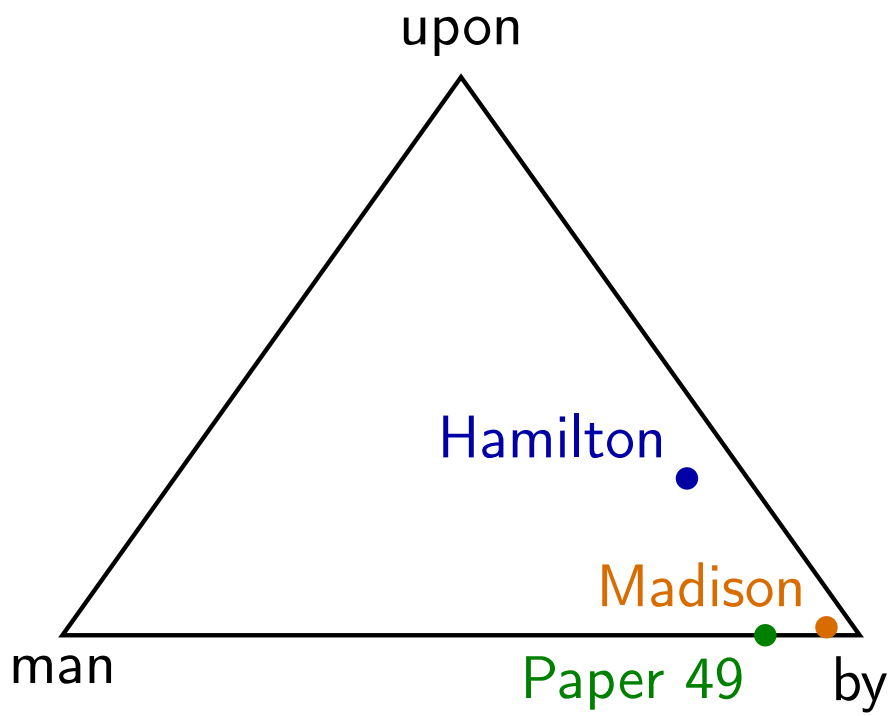
$$\mathbf{x}_{49} = (0.117647,\ 0.882353,\ 0).$$

Cosine similarity compares direction and ignores overall length:

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}.$$

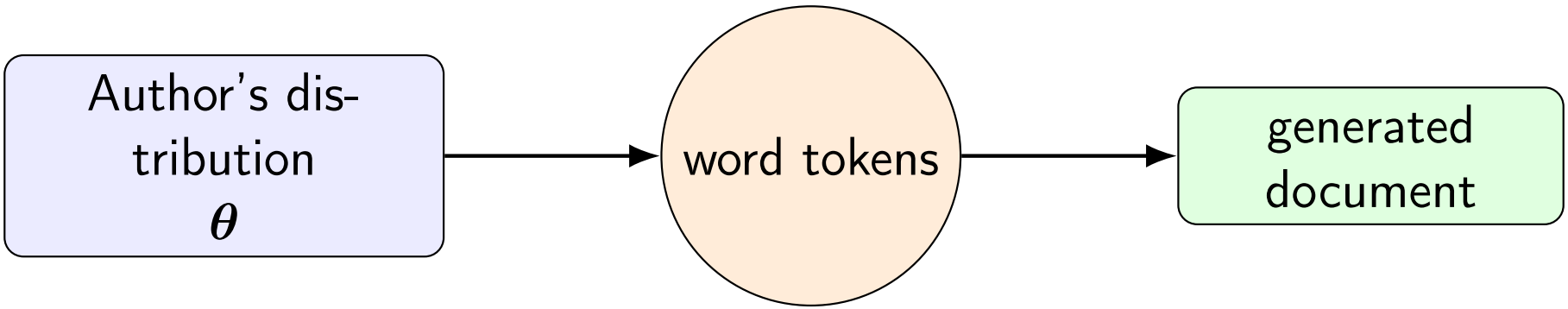
For paper 49, cosine similarity is 0.918 to Hamilton and 0.995 to Madison.

Three-word proportions live in a simplex



Each point conditions on the selected words; the three proportions sum to one.

A multinomial model of language



The model says that an author's word distribution generates the observed counts. We compare how well Hamilton's and Madison's distributions explain a target paper.

The geometry has a denominator

There are two legitimate but different vectors:

- all-token rates: each word count divided by the essay's total tokens;
- conditional three-word rates: each count divided by the total count of *man*, *by*, and *upon*.

Do not compare one denominator to the other without saying what the vector means.

A small multinomial model

Let c_w be the count of selected word w in a paper and $N = \sum_w c_w$.

$$P(\mathbf{c} \mid \boldsymbol{\theta}) = \frac{N!}{\prod_w c_w!} \prod_w \theta_w^{c_w}.$$

For the selected vocabulary $\{\text{man}, \text{by}, \text{upon}\}$, N is the count of those three words, not the essay's full token count.

Smoothing keeps zeroes from ending the comparison

With additive smoothing $\alpha = 1$ and $K = 3$ selected words,

$$\hat{\theta}_w = \frac{C_w + \alpha}{N + \alpha K}.$$

For the reference profiles:

	man	by	upon
Hamilton	.078358	.641791	.279851
Madison	.035928	.948104	.015968

Smoothing is a modeling choice. It does not create evidence that was absent.

Compare log likelihoods, not tiny probabilities

The multinomial coefficient is common to both authors for one target paper, so it cancels in a likelihood comparison:

$$\ell(a) = \sum_{w} c_w \log \hat{\theta}_{w,a}.$$

For paper 49, counts are $(2, 15, 0)$ and $\ell_H - \ell_M = -4.29$ with $\alpha = 1$. The selected-word model therefore favors Madison.

Classify the eleven source-labelled disputed papers

Paper	49	50	51	52	53	54	55	56
H–M log likelihood	-4.29	-1.43	-8.20	-8.58	-9.76	-3.64	-5.46	-3.90

Paper	57	62	63
H–M log likelihood	-8.20	-10.15	-18.34

All eleven source-labelled disputed papers are closer to Madison under this model.

A check against known essays

Fit the same three-word model repeatedly, leaving one known Hamilton or Madison paper out of its reference set.

The leave-one-out rule gets 64 of 66 known H/M essays correct with this tokenization and smoothing choice.

That is a useful diagnostic, not proof that the model will classify the disputed papers correctly: the disputed set may differ in topic, editing, or historical status.

Log odds across the vocabulary

For every word in the H/M vocabulary, compare smoothed rates:

$$\log \frac{\hat{p}(w \mid H)}{\hat{p}(w \mid M)} = \log \frac{C_{w,H} + 1}{N_H + K} - \log \frac{C_{w,M} + 1}{N_M + K}.$$

The largest positive value is *upon* (2.94); the most negative is *democracy* (-3.30).

Rare words can produce extreme values. A frequency threshold is a sensitivity check, not an afterthought.

Counts, TF–IDF, and retrieval

The selected-word model asks: which author profile gives this paper the larger likelihood?

TF–IDF asks a different question: which words make this paper distinctive within a collection?

$$\mathrm{tfidf}(w, d) = \mathrm{tf}(w, d) \log \frac{D}{\mathrm{df}(w)}.$$

A nearest-neighbour search can retrieve thematically similar essays, but topical similarity is not the same as authorship.

Sensitivity is part of the result

Change one decision at a time and record the consequence:

- keep or remove the shared greeting and signature;
- function words only versus a mixed vocabulary;
- raw counts versus rates;
- $\alpha = 0.1$, 1, or another justified value;
- selected words versus a documented frequency-filtered vocabulary.

If the classification changes, the disagreement is about the measurement and model, not a mysterious property of the essays.

What this example teaches

- A historical question becomes a sequence of measurement decisions
- Preprocessing is part of the construct, not simply a cleanup
- A count, rate, vector, likelihood, and log odds answer different questions
- Held-out checks expose whether a simple rule travels beyond its training examples
- A numerical classification should remain conditional on data, features, and model

Sources and further reading

- Library of Congress, *The Federalist Papers*: guides.loc.gov/federalist-papers
- Mosteller, Frederick, and David L. Wallace. 1963. "Inference in an Authorship Problem." *JASA* 58(302):275–309.
- Mosteller and Wallace. 1964. *Inference and Disputed Authorship: The Federalist*.
- Grimmer, Justin, Margaret E. Roberts, and Brandon M. Stewart. 2022. *Text as Data*, chapters 2–3.